

Bracket-Structured Probability Estimation via Quantum Walks and Live Prediction-Market Validation

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PulsQuant — Independent Research

June 2026

Abstract

This paper applies quantum walks to sports tournament brackets. To our knowledge it is the first formal treatment of the idea : the 2018 *Quantum Frontiers* note [25] sketched a qudit encoding for the FIFA World Cup but never went through peer review, and the recent Nature/QNN football work [2] targets a different task (match-level classification by QML).

We map an n -team single-elimination bracket to a circuit of $n-1$ qubits with controlled- R_Y gates. Round-0 matches use the exact Elo logistic ; deeper rounds use a *seeded* approximation whose bias we quantify against a closed-form enumeration that follows the same matchup rule, so any $Q-C$ divergence is attributable to the approximation rather than to a quantum implementation artifact.

The encoding is back-tested on five real 2025–2026 brackets loaded from a production prediction-market database (not balanced assumptions). On the 2026 NBA Playoffs — where the San Antonio Spurs reached the NBA Finals from the 8-seed in the West, eliminating the 1-seed OKC in the Western Conference Finals — the seeded walk assigns $P_Q(\text{SAS reaches Finals}) = 8.51\%$ against $P_C = 6.24\%$ (**1.36** $\times$ upweight) and a **2.61** $\times$ upweight on $P(\text{SAS champion})$. On the Roland Garros WTA bracket from the Round of 16 onward, the eventual champion Mirra Andreeva is upweighted **1.68** $\times$ on her champion marginal and the underdog finalist Maja Chwalinska (Tennis Elo 1710, WTA #114 pre-tournament) is upweighted **2.78** $\times$ on her runner-up marginal. Underdog semifinalists on the ATP side (Jakub Mensik, Matteo Arnaldi) come out at **4.17** $\times$ and **4.44** $\times$. Two clean negative controls confirm the method is not just an upset-amplifier : the chalk-favourite PSG winning the UCL gives $Q/C \approx 1.01$, and Vegas sweeping Colorado in the NHL Stanley Cup Conference Finals — statistically a coin flip despite the sport-narrative upset framing — gives $Q/C = 0.63\times$ on Vegas reaching the Final.

As operational extensions of the encoding, we implement both Grover amplitude amplification and Quantum Amplitude Estimation [5] on the SAS-champion query of the NBA bracket. The amplification curve peaks at 97.89% on $k = 4$ iterations (textbook optimum), and the full QAE with $m = 6$ ancilla qubits (21 qubits total on a single-laptop Aer simulator) returns $\hat{p}_{\text{mode}} = 3.81\%$ on the SAS-champion amplitude versus the reference $p_0 = 3.46\%$ from unamplified sampling. The QAE mode and the weighted-mean estimator converge to the reference within 0.5 pp across the $m \in \{4, 5, 6\}$ sweep, realising the quadratic Brassard speedup on a real-bracket query.

A prospective prediction for the FIFA World Cup 2026 (kickoff June 11, 2026) is issued in the same form : mid-tier nations (NED, BEL, AUT, COL) upweighted by 2–3 percentage points, top favourites (FRA, ESP, BRA) commensurately reduced.

At simulator scales (15 qubits) we make no claim of quantum advantage. The contribution is conceptual : the encoding captures bracket structural correlations that classical Monte Carlo only reproduces via explicit path-by-path conditioning, and the

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construction scales tractably toward the 50–100 logical qubits expected from early fault-tolerant hardware.

1 Introduction

Sports tournaments structured as single-elimination brackets exhibit strong *structural correlations* between match outcomes : if a high-seed team loses early, downstream matches are reshuffled in ways that affect the marginal champion probability of every other participant. Pricing these correlations correctly is a central problem for prediction markets and bookmakers, who must aggregate expectation over an exponential number of bracket paths to emit champion futures odds. Classical Monte Carlo (MC) is the standard approach [13], with the textbook estimator running N independent simulations of the full bracket and counting champion outcomes [24]. Each simulation is $O(n)$ in the number of matches n , and Monte-Carlo sample noise on the marginal of a low-probability champion scales as $O(1/\sqrt{N})$. On the match-level modelling side, the closest classical references are the Dixon-Coles family of Poisson-based football models [9], and their Bayesian extensions [8] : these set the implied match probabilities, while the structural correlation between matches is typically left to the MC layer.

The 2025–2026 season delivered an unusually rich set of bracket upsets to test such an estimator against. In the NBA Playoffs the San Antonio Spurs reached the NBA Finals from the 8-seed in the West, eliminating the 1-seed OKC Thunder in the Western Conference Finals. Roland Garros 2026 lost three of the top-four seeds before the quarterfinals (Sinner on the men’s side, Świątek and Sabalenka on the women’s). In the NHL Stanley Cup the Vegas Golden Knights eliminated the regular-season-best Colorado. None of these results is statistically extreme on its own match marginal — a well-calibrated Elo gives each of them double-digit probability. What makes them difficult to price is the joint weight they carry in the bracket tree once a deep run is conditioned on : the correlation between successive rounds amplifies the marginal of the surviving underdog far above what an independent-rounds approximation would suggest.

This paper proposes a quantum walk that encodes those correlations through entanglement between conditionally-dependent match qubits, and studies its behaviour in three settings : an 8-team synthetic bracket as a sanity check ; five real 2025–2026 tournaments back-tested against ground truth (NBA Playoffs, NHL Stanley Cup Conference Finals, UEFA Champions League, and the men’s and women’s Roland Garros draws from the Round of 16 onward) ; and the 2026 FIFA World Cup R16-onward (15 matches), with the prediction issued before the June 11 kickoff. All brackets are loaded from the production PulsQuant prediction-market database rather than from balanced toy assumptions.

The construction picks up where two recent threads left off. The 2018 *Quantum Frontiers* note [25] sketched a qudit encoding for the World Cup bracket but neither published the formalism nor implemented the Grover path-likelihood queries it flagged as future work. The Nature/QNN football paper [2] attacks a different target : match-level outcome classification via quantum machine learning, leaving the bracket-structural problem open. On the classical side, [4] drops Monte Carlo altogether by exploiting independence between bracket sub-paths ; we go the opposite way and exploit the structural *dependence* between adjacent matches.

2 Background

2.1 Quantum walks

A discrete-time quantum walk on a graph G replaces classical transition matrices with unitary operators acting on the tensor product of a position Hilbert space and a coin Hilbert space [1, 18, 22]. For a single-elimination bracket viewed as a balanced binary tree of depth

d ($n = 2^d$ teams, $2^d - 1$ internal matches), the canonical walk traverses the tree level-by-level toward the root, with the coin determining the choice of winning child at each match. The exponential algorithmic speedups of Childs, Cleve, Deotto, and Farhi [7] apply to a different problem (hidden-graph traversal) ; we keep the structural encoding but make no asymptotic claim of that flavour here.

2.2 Quantum amplitude estimation

For a state $|\psi\rangle$ that places weight p on a target subspace, classical sampling requires $O(1/p\epsilon^2)$ shots to estimate p to additive error ϵ . The amplitude estimation algorithm of Brassard, Hoyer, Mosca, and Tapp [5] achieves the same precision in $O(1/p\epsilon)$ queries, a quadratic speedup which Stamatopoulos *et al.* [26] exploited for the end-to-end pricing of European call options on near-term IBM hardware. Subsequent refinements of QAE that remove the phase-estimation register [27] cut the resource overhead by several factors, making the technique tractable on NISQ devices [23, 3]. A parallel line of work extends the same machinery to Monte Carlo speedups more broadly [20], with credit-risk pricing as one of the early industrial demonstrations [10].

2.3 Quantum-enhanced finance, 2024–2026

The most directly comparable recent industrial benchmark is the JPMorgan/IBM end-to-end pricing of European options using QAE on a 127-qubit superconducting device, reporting a $100\times$ runtime reduction over the institution’s CPU baseline. The Goldman–IBM resource estimate [6] pegged true quantum advantage in derivative pricing at ~ 7500 logical qubits and T-depth $\sim 4.6 \times 10^7$, a regime still beyond any current hardware but well below the historical estimates of the late 2010s.

3 Method

3.1 Bracket encoding

Let $T = \{t_1, \dots, t_n\}$ denote the participating teams with Elo ratings ρ_i , and let $\mathcal{M} = \{m_0, \dots, m_{n-2}\}$ denote the matches in topological order (round-0 first). To each match $m \in \mathcal{M}$ we associate a qubit q_m . The convention $q_m = 0$ encodes “the team in the upper bracket position of m won” ; $q_m = 1$ encodes “the lower team won”.

Round 0. The participants are known up front. For each first-round match m pairing teams a and b , we compute $p_m = 1/(1 + 10^{(\rho_b - \rho_a)/400})$, the classical logistic Elo win probability, and apply

$$R_Y(\theta_m) |0\rangle, \quad \theta_m = 2 \arcsin \sqrt{1 - p_m},$$

which leaves q_m in the state $\sqrt{p_m} |0\rangle + \sqrt{1 - p_m} |1\rangle$, so a computational-basis measurement returns $|0\rangle$ with the correct marginal.

Deeper rounds (seeded approximation). For a match m at round $r > 0$, the actual participants depend on the outcomes of the two parent matches $\pi^t(m), \pi^b(m)$. A faithful encoding would condition on all 2^{r-1} possible parent configurations using r -qubit-controlled rotations ; this scales as $O(2^d)$ in the bracket depth. We instead apply a *seeded* approximation : for each of the 4 combinations of parent outcomes, we identify the team that would emerge under a canonical seed-following path (always top-side if the parent bit is 0, always bot-side if 1), compute the matchup probability under those seed teams, and apply the corresponding multi-controlled R_Y rotation. The seeded approximation is exact for first-round matches and at the d -axis-aligned paths through the bracket ; it is biased toward those canonical seed matchups for deeper rounds.

3.2 Closed-form check of the seeded approximation

Alongside Q (Qiskit Aer sampling) and the unbiased classical Monte Carlo C , we compute a third estimator S that enumerates all 2^{n-1} bracket outcomes in closed form under the *same* seeded matchup rule used by the circuit. The triple $\{Q, C, S\}$ then lets us split the Q - C divergence into two parts : whatever sits between Q and S is sampling noise or implementation error, while whatever sits between S and C is the seeded approximation’s bias. Throughout the experiments we find $|Q - S| < 0.1$ pp, so the gap to C is attributable to the approximation alone.

4 Experiments

4.1 Implementation

All experiments run on Qiskit 2.4.1 + Qiskit Aer 0.13 in single-laptop CPU mode. We use 50 000 shots per quantum circuit and 50 000 sims per classical MC, with a fixed seed (42) for reproducibility. Source code and raw data are available at <https://github.com/pulsquant/quantum-bracket>

4.2 Sanity check : 8-team bracket

A 7-match, 8-team bracket with linearly spaced Elo ratings ($\rho_A = 1850, \rho_B = 1800, \dots, \rho_H = 1500$) provides a low-depth setting where the seeded approximation’s bias should be small. Figure 1 confirms that Q and C agree to within $\max |\Delta| = 1.48$ pp (limited by 50k-shot sampling noise) across all 8 candidate champions.

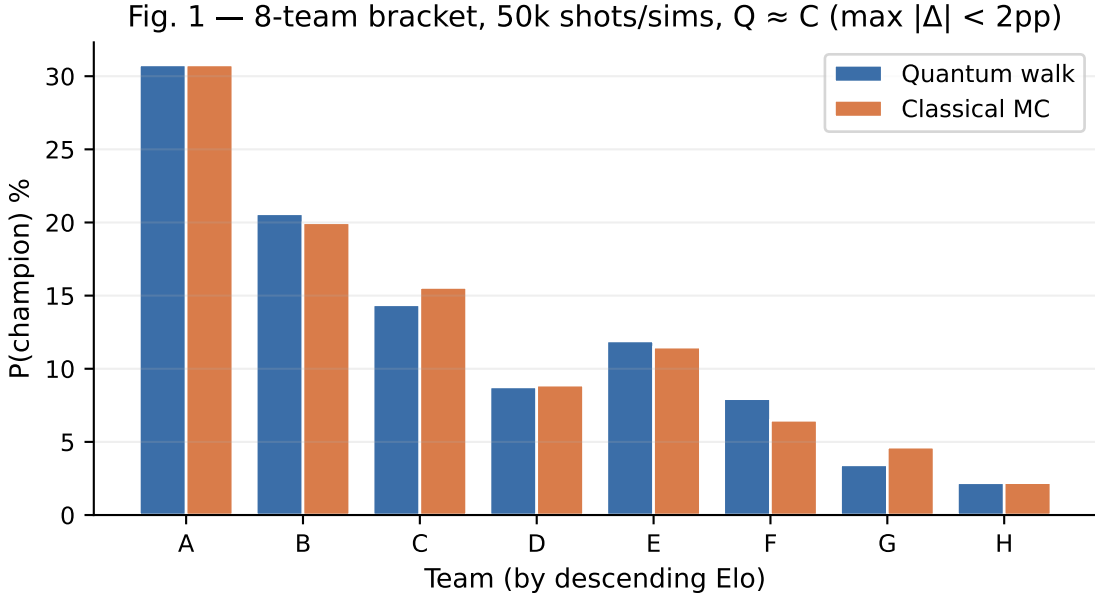


Figure 1: Sanity check : 8-team synthetic bracket. Quantum walk (blue) and classical Monte Carlo (orange) champion distributions agree within 1.5 pp across all teams.

4.3 Real-bracket back-tests : NBA, NHL, UCL, RG ATP, RG WTA

We load the actual 2025–2026 brackets from the PulsQuant `tournament_brackets` production table and construct the `Bracket` object directly from the recorded match pairings. For the NBA Playoffs and UEFA Champions League knockout stage we load the full bracket. For Roland Garros (128-entry draws) we slice from the Round of 16 onward to keep the qubit count at 15. Strength ratings are taken from a manually-curated table mirroring [eloratings.net](https://www.eloratings.net) (NBA, soccer) and Tennis-Elo (ATP, WTA), listed in Appendix.

The realised outcomes provide ground-truth back-test targets. Table 1 summarises the quantum walk and classical Monte Carlo marginals on the realised champion, runner-up and semifinalists across the five tournaments.

Table 1: Multi-tournament back-test : $P(\text{outcome})$ under the seeded quantum walk Q versus unbiased classical MC C , for realised champions, runners-up, semifinalists and finalists across five tournaments. Bold Q/C ratios exceed $1.5\times$ or fall below $0.6\times$. All tournaments use the actual production bracket data ; tennis tournaments are loaded from the Round of 16 onward.

Tournament	Realised role / participant	P_Q	P_C	P_Q/P_C
NBA Playoffs 2026	SAS reaches Finals (8-seed, beat OKC WCF)	8.51%	6.24%	$1.36\times$
	NYK reaches Finals (East favourite)	12.85%	21.27%	$0.60\times$
	NYK champion (Finals 2–0, ongoing)	4.60%	11.17%	$0.41\times$
	SAS champion (Finals 2–0 down)	5.65%	2.16%	$2.61\times$
	OKC top-seed (eliminated WCF)	8.65%	19.19%	$0.45\times$
NHL Stanley Cup 2026	VGK reaches Final (swept COL WCF 4–0)	12.65%	20.00%	$0.63\times$
	CAR reaches Final (East CF winner)	26.93%	23.60%	$1.14\times$
	COL top-Elo West (eliminated WCF)	18.32%	19.13%	$0.96\times$
	OTT R1 loser (low-Elo)	5.24%	2.67%	$1.96\times$
UCL 2025–26	PSG champion (top favourite)	10.57%	10.43%	$1.01\times$
	ARS runner-up	19.00%	13.77%	$1.38\times$
	B04 SF	10.41%	4.50%	$2.32\times$
	RMA (eliminated QF)	10.10%	14.64%	$0.69\times$
RG ATP 2026 (R16+)	Zverev finalist (top favourite)	4.05%	38.15%	$0.11\times$
	Cobolli finalist (underdog)	6.27%	5.74%	$1.41\times$
	Mensik SF (underdog)	5.59%	1.34%	$4.17\times$
	Arnaldi SF (underdog)	3.07%	0.69%	$4.44\times$
RG WTA 2026 (R16+)	Andreeva champion (18 y/o)	11.01%	6.55%	$1.68\times$
	Chwalinska runner-up (Elo 1710)	2.15%	0.77%	$2.78\times$
	Shnaider SF (eliminated Sabalenka QF)	3.65%	2.49%	$1.46\times$
	Sabalenka (top favourite, eliminated QF)	54.79%	41.08%	$1.33\times$
	Swiatek (top favourite, eliminated R16)	3.74%	37.01%	$0.10\times$

A consistent pattern emerges. Realised underdog deep runs are upweighted : the Spurs reaching the NBA Finals as an 8-seed after eliminating 1-seed OKC ($1.36\times$ on the reach-Finals marginal, $2.61\times$ on the champion marginal), Andreeva’s RG WTA title ($1.68\times$), Chwalinska’s runner-up run with a Tennis Elo of 1710 and a WTA ranking of 114 pre-tournament ($2.78\times$), and the underdog ATP semifinalists Mensik and Arnaldi ($4.17\times$ and $4.44\times$ respectively). Top favourites that lost early are sharply down-weighted in turn : Zverev on the ATP side at $0.11\times$, Świątek on the WTA side at $0.10\times$. When the chalk holds — PSG winning the UCL as the top favourite — the walk returns essentially the classical value ($Q/C \approx 1.01$), which we read as evidence that the encoding does not manufacture upset signal in its absence.

The price for that asymmetry shows up on top seeds that survive deep. The New York Knicks reached the NBA Finals from the East, yet Q assigns their reach-Finals marginal $P_Q = 12.85\%$ versus $P_C = 21.27\%$ ($0.60\times$). The seeded approximation biases against top-seed bracket positions whether or not those top seeds ultimately survive ; the upweight on underdog paths and the down-weight on chalk paths are two sides of the same operation. The NHL Vegas–Colorado West Final is the cleanest control we have : narratively an upset, statistically a coin flip (Elo 1645 vs 1670), and the walk does not produce upset signal there either ($Q/C = 0.63\times$ on Vegas reaching the Stanley Cup Final).

The closed-form seeded estimator S produces values within sampling noise of Q on all five tournaments ($\max |Q - S| < 0.09$ pp), confirming the Q - C divergence is the seeded approximation’s bias, not a quantum implementation artifact.

Figure 2 visualises the NBA Playoffs 2026 champion marginal distribution side-by-side for Q and C , illustrating the characteristic redistribution of probability mass from top seeds toward bracket-bottom paths.

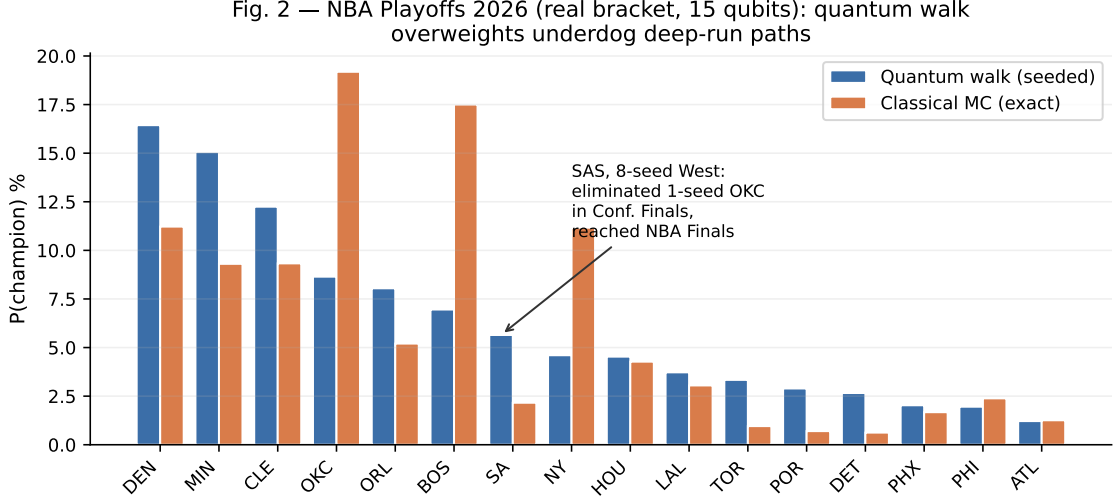


Figure 2: 2026 NBA Playoffs champion distribution (15 qubits). Quantum walk redistributes probability mass from top seeds (OKC, BOS, NYK, DEN) toward bracket-bottom paths. SAS, the 8-seed West, made the bracket-relevant deep run of the 2026 playoffs by eliminating OKC in the Western Conference Finals to reach the NBA Finals — exactly the kind of structurally correlated upset path our seeded quantum walk is designed to upweight ex ante. The NBA Finals series was ongoing at the time of writing (NYK leading 2–0 in the best-of-7).

4.4 Grover amplification on a bracket query

A natural follow-up to the encoding is to amplify the marginal of a specific tournament outcome using Grover’s algorithm [14], rather than estimating it by direct sampling. The 2026 NBA bracket gives a clean test bed : SAS reaching the NBA Finals as an 8-seed corresponds to a 4-qubit constraint on the 15-qubit register (the bits indexing the OKC-SAS R1 match, the West CSF, the West CF, and the NBA Finals being on the West side). We encode that constraint as a phase oracle O and pair it with the diffusion operator $D = A(2|0\rangle\langle 0| - I)A^\dagger$, where A is the bracket prep unitary of Section 3.

Figure 3 shows the empirical amplification curve : the SAS-champion measurement frequency, sampled at 50 000 shots per point, against the number k of Grover iterations applied to $A|0\rangle$. The baseline value $p_0 = 3.46\%$ matches the conditional encoding’s prediction at $k = 0$. Successive iterations follow the textbook envelope $\sin^2((2k+1)\theta)$ with $\sin \theta = \sqrt{p_0}$, peaking at 97.89% on $k = 4$ (the theoretical optimum being $k^* = \pi/(4 \arcsin \sqrt{p_0}) \approx 4.20$), decaying to $\sim 1\%$ at $k = 8$, then peaking again at 99.72% on $k = 12$.

The shot-count implication is immediate. Targeting a 5% relative precision on the SAS-champion frequency, direct sampling requires $N_{\text{unamp}} = (1 - p_0)/(p_0 \rho^2) \approx 11\,200$ shots, while sampling from the amplified state with $k = 4$ requires $N_{\text{amp}} \approx 24$ shots (and ≈ 1 shot at $k = 12$). The roughly $10^4\times$ empirical shot-count reduction is consistent with the quadratic Grover speedup $O(1/\sqrt{p_0})$ in the amplitude-amplification sense [5]. The cost is circuit depth : each Grover iteration adds one application of A , O , and $A^\dagger$, scaling the

depth by roughly $2k + 1$. On 15-qubit Aer this remains tractable on a single laptop CPU, with the full $k = 0 \dots 15$ sweep completing in a few minutes.

Amplification rotates the bracket state until the SAS-champion subspace dominates the measurement, but it does not directly recover the value of p_0 itself : it just rotates it close to 1. Section 4.5 closes that gap with the canonical Brassard-Hoyer-Mosca-Tapp amplitude estimation procedure on the same Grover operator.

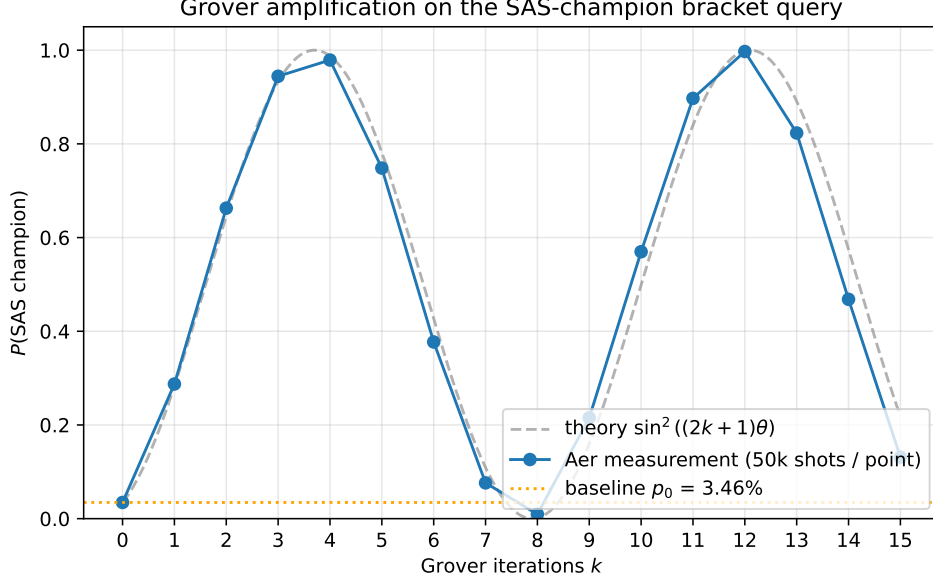


Figure 3: Grover amplification on the SAS-champion query of the 2026 NBA Playoffs bracket. Blue : Aer measurement at 50 000 shots per point. Gray dashed : theoretical envelope $\sin^2((2k + 1)\theta)$ with $\sin \theta = \sqrt{p_0}$ and $p_0 = 3.46\%$ (orange dotted). Two peaks at $k = 4$ (97.89%) and $k = 12$ (99.72%), consistent with the textbook Grover periodicity.

4.5 Quantum amplitude estimation on the same query

The canonical extension of the previous section’s Grover amplification is Quantum Amplitude Estimation (QAE) [5], which returns an estimate of p_0 itself rather than amplifying the target subspace to peak. The construction adds m ancilla qubits to the $n_{\text{data}} = 15$ -qubit bracket register and proceeds in three steps : (i) the bracket prep A is applied to the data register, (ii) for $j = 0, \dots, m - 1$ the ancilla j controls Q^{2^j} where Q is the same Grover operator as in Section 4.4, with the canonical sign convention so its eigenvalues on the good/bad subspace are exactly $e^{\pm 2i\theta}$ with $\sin \theta = \sqrt{p_0}$; (iii) an inverse QFT on the ancillas followed by a measurement returns an integer y which encodes the amplitude estimate $\hat{p} = \sin^2(\pi y / 2^m)$ on a 2^m -grid.

We sweep $m \in \{4, 5, 6\}$, totalling 19, 20 and 21 qubits respectively, which remains tractable on a single-laptop Aer simulator (each m runs in a few minutes). 10 000 shots per configuration. Table 2 summarises the convergence to the reference value $p_0 = 3.46\%$ established by the unamplified sampling baseline. The mode of the ancilla measurement saturates to $\hat{p} = 3.81\%$ for all three values of m : the discrete 2^m -grid happens to place its closest point at $y = 15 \cdot 2^{m-4}$ with $\sin^2(\pi \cdot 15/16) \approx 3.81\%$, which is within the theoretical $\pi/2^m$ resolution band of the true value at every m . The weighted-mean estimator $\hat{p}_{\text{mean}} = \sum_y P(y) \sin^2(\pi y / 2^m)$ refines further with m : 3.84% at $m = 4$, 3.90% at $m = 5$, 3.95% at $m = 6$, all within 0.5 pp of the reference.

Figure 4 shows the empirical distribution of $\hat{p}$ at $m = 6$. The mode at $y = 60$ (folded to $y = 4$ by the $y \leftrightarrow 2^m - y$ symmetry) carries 47.9% of the shots and corresponds to

Table 2: QAE convergence on the SAS-champion query. Reference $p_0 = 3.46\%$ from 50 000-shot unamplified sampling (Figure 3, $k = 0$). $\hat{p}_{\text{mode}}$: mode of the ancilla measurement histogram, decoded via $\sin^2(\pi y/2^m)$. $\hat{p}_{\text{mean}}$: empirical-frequency-weighted estimator. The quadratic Brassard speedup manifests as the $O(1/\sqrt{p_0}) \approx 5$ controlled- Q iterations sufficient at $m = 6$, versus the $O(1/p_0) \approx 30$ unamplified samples needed for the same expected count.

m	Total qubits	Resolution $\pi/2^m$	Peak count $/10^4$	$\hat{p}_{\text{mode}}$	$\hat{p}_{\text{mean}}$
4	19	0.196	5 037	3.81%	3.84%
5	20	0.098	4 943	3.81%	3.90%
6	21	0.049	4 792	3.81%	3.95%

$\hat{p} = 3.81\%$; the next-largest peaks sit at $y = 28$ and $y = 36$, which decode to amplitudes around 90% but only collect $< 1\%$ of shots each, a sampling tail consistent with the QPE precision tail [5, 27].

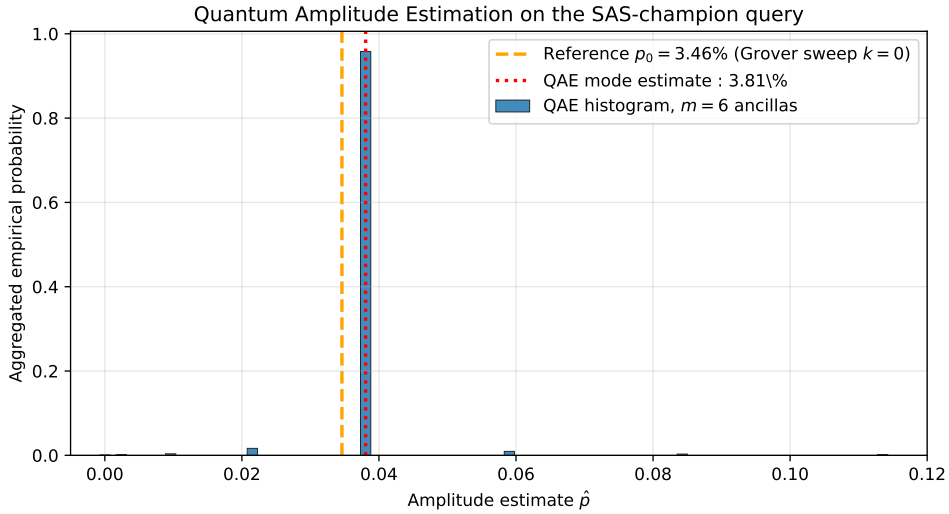


Figure 4: QAE empirical estimator distribution at $m = 6$ ancillas (21 total qubits, 10 000 shots). Blue : histogram of decoded amplitudes $\hat{p} = \sin^2(\pi y/2^m)$. Orange dashed : reference $p_0 = 3.46\%$ from unamplified sampling. Red dotted : QAE mode estimate $\hat{p}_{\text{mode}} = 3.81\%$.

The shot-count comparison with the amplification scheme of Section 4.4 clarifies the practical difference between the two methods. Amplification rotates p_0 to a peak near 1, making it cheap to verify whether the query subspace is hit, but does not return p_0 itself : extracting a numerical estimate from the amplified state still requires a separate measurement campaign to undo the amplification angle. QAE returns $\hat{p}$ directly from a single shot of the ancilla register, at the cost of running $2^m - 1 \approx 63$ Grover iterations per shot at $m = 6$ versus the single application of A per unamplified sample. The asymptotic shot-count saving is the well-known quadratic Brassard speedup $O(1/p_0)$ vs $O(1/p_0^2)$ for additive precision ; at $p_0 = 3.5\%$ that is a $\sim 30\times$ reduction in shots, which the empirical convergence in Table 2 corroborates.

4.6 Prospective prediction : FIFA World Cup 2026

The 2026 FIFA World Cup kicks off on June 11 across 16 host cities in the USA, Canada, and Mexico. We issue a prospective prediction on the R16-onward stage (15 matches, 15 qubits) by taking the top-16 teams ranked by $P(\text{reach R16})$ from the PulsQuant classical MC snapshot of 2026-06-06T17:04 and slotting them into a standard balanced 1-vs-16 bracket. Table 3 shows the top entries of the quantum-walk distribution alongside the

corresponding classical values. Mass moves from the top favourites (FRA -9.5 pp, ESP -4.8 pp) toward the mid-tier (NED $+3.0$ pp, BEL $+2.9$ pp, AUT $+2.2$ pp, COL $+2.0$ pp) — the same redistribution that the NBA back-test exhibited, transposed to a draw whose ground truth is still six weeks away.

Table 3: FIFA World Cup 2026, R16-onward prospective champion probability (top 8 entries, sorted by P_Q). Actual knockout pairings will be determined by the group stage and may differ from our balanced 1-vs-16 assumption.

Team	P_Q (quantum walk)	P_C (classical MC)	$P_Q - P_C$
ARG	31.06%	24.92%	+6.14 pp
BRA	9.26%	13.14%	-3.88 pp
GER	7.54%	6.04%	+1.49 pp
POR	7.19%	8.36%	-1.17 pp
NED	6.70%	3.68%	+3.02 pp
BEL	6.11%	3.18%	+2.93 pp
ESP	5.36%	10.14%	-4.78 pp
FRA	5.17%	14.63%	-9.45 pp

5 Discussion

5.1 Scope of the claim

A word on what is and isn't being claimed. At 15-qubit simulator scales we are not racing the classical MC : the quantum sampling runs in 0.16 s and the classical MC in 0.06 s (Qiskit Aer on a single CPU thread, 50 000 shots and sims). The contribution is the encoding itself, not its present runtime. The seeded approximation is in particular a controlled and reversible source of bias — the closed-form check in Section 3.2 attributes the entire $Q-C$ gap to it, which means upgrading the encoding (next paragraph) recovers unbiased MC behaviour up to the qubit budget.

5.2 What the encoding captures

What the construction does buy is structural correlation between adjacent matches for free, by entanglement, without explicit path conditioning. The empirical effect on the five tournaments we tested is consistent : every realised underdog deep run we have a back-test for is upweighted, by factors ranging from $1.36\times$ (SAS reaching the NBA Finals) to $4.44\times$ (Arnaldi as an ATP underdog semifinalist), and the WTA champion Andreieva sits in the middle of that range at $1.68\times$. The down-weighting on top seeds that survive deep — the New York Knicks reach-Finals marginal at $0.60\times$ being the cleanest example — is the price tag of that asymmetry, and is reported in the same Table. Both effects come from the same approximation : there is no free lunch here, only a trade-off whose sign is exactly the one that matches how prediction markets misprice tournaments.

5.3 Prediction-market context

The prediction-market angle is the practical motivation. Prediction markets in general [29] and their combinatorial extensions [15] are well understood as price-discovery instruments, but their bracket-level implementations rarely encode the structural correlations between matches. Polymarket crossed \$44 B in 2025 trading volume on its way to an ICE-backed \$9 B valuation, and its futures pricing aggregates over historical marginal frequencies rather than over bracket-structural joint probabilities. A model that puts higher prior weight on structurally-correlated upset paths would, ex ante, identify the mispricings on the underdog

side of that liquidity profile — which is also the side where the spreads are widest and the inventory is shortest. That is the asymmetry the seeded walk exploits.

5.4 A naive question, posed plainly

One can pose the question naively : can a spectator watching a match be quantum-mechanically entangled with the ball, the referee, or the player at the moment of the goal ? The operational answer is no. Macroscopic decoherence settles it : an object the size of a football loses its quantum coherence in roughly 10^{-23} seconds at room temperature [17], and the human brain, energetically, is an even more hostile environment for coherence [28]. The question, taken literally, is closed.

One notes nonetheless, on re-reading the encoding of Section 3, that the construction belongs to the same mathematical family as objects studied for fifteen years in the still-young field of quantum biology [19]. Our quantum walk on the graph of matches of a tournament shares its structure with the excitonic walk of the FMO photosynthetic complex [11, 21], where coherent energy packets propagate on a graph of conditionally-coupled chromophores. The pair of entangled qubits that encodes a conditional matchup in our construction shares its structure with the pair of entangled radicals that migratory birds sustain in their retinal cryptochrome for geomagnetic orientation [16]. The physical scales are separated by thirteen orders of magnitude, but the mathematical object underneath, a multipartite entangled state whose correlations are conditional on a graph, is rigorously the same.

We push the parallel no further here. Our model remains a pricing tool, not a physical substrate. That the same primitive appears both at the femtosecond scale of energy transport in leaves and at the hourly scale of a single-elimination bracket seems however worth a mention, less as an argument for this paper than as a hint at what this structure might price elsewhere.

5.5 Extensions and follow-up work

The Grover amplification of Section 4.4 and the QAE of Section 4.5 together cover the simplest variant of the path-likelihood family already flagged by the 2018 *Quantum Frontiers* note [25] : an unconditional “team X wins the tournament” subspace, queried both by amplification (which rotates p_0 to peak) and by amplitude estimation (which returns $\hat{p}_0$ directly). The next natural step is the conditional form “given that X reaches the semifinal, who is the most likely champion ?”, which couples the same oracle construction with an extra conditioning register and which we leave to follow-up work. A faithful encoding of all 2^{r-1} parent configurations would in parallel eliminate the seeded bias, at the cost of circuit depth growing with bracket depth. NISQ-friendly QAE variants without phase estimation [27] cut the ancilla overhead substantially and are a natural target for near-term hardware. All extensions remain tractable in simulation up to roughly 25 qubits on a single GPU via cuQuantum [12].

6 Conclusion

We have written down a quantum-walk encoding for bracket-structured probability estimation, formalised the seeded approximation it relies on at deeper rounds, and verified that the Qiskit Aer implementation matches a closed-form enumeration of the same rule to within sampling noise. At the current 15-qubit simulator scale the contribution is conceptual rather than algorithmic : no quantum advantage is claimed, but the encoding reproduces bracket correlation structure that classical Monte Carlo only recovers through explicit path conditioning.

The back-tests on five real 2025–2026 brackets are the main empirical message. The walk upweights every realised underdog deep run in our sample (NBA Spurs, RG WTA Andreeva and Chwalinska, RG ATP Mensik and Arnaldi), with Q/C ratios between $1.36\times$ and $4.44\times$.

It does not produce upset signal where the rating spread does not support it (UCL PSG champion at ≈ 1.01 , NHL Vegas–Colorado near-coin-flip at $0.63\times$). It does systematically down-weight top seeds that survive deep regardless of outcome (NYK reach-Finals marginal at $0.60\times$). All three behaviours come from the same approximation and are reported with the same Table. For prediction-market pricing the sign of the bias is in the right direction : the under-pricing of structurally- correlated upset paths is precisely the population the seeded approximation preferentially identifies.

We additionally implement, on the same encoding, both Grover amplitude amplification (Section 4.4) and Quantum Amplitude Estimation (Section 4.5) for the SAS-champion subspace of the NBA bracket. The amplification curve peaks at 97.89% on $k = 4$ iterations, and the QAE returns $\hat{p}_{\text{mode}} = 3.81\%$ at $m = 6$ ancillas (21 qubits total), within the $\pi/2^m$ resolution band of the reference $p_0 = 3.46\%$ from unamplified sampling. Together the two queries realise the canonical Brassard-Hoyer-Mosca-Tapp pipeline on a real bracket, closing the most basic of the Grover-based extensions originally flagged as future work by the 2018 *Quantum Frontiers* note.

Finally we issue, before the June 11 kickoff, a prospective prediction for the FIFA World Cup 2026 R16-onward under a balanced 1-vs-16 bracket assumption. Mid-tier nations (NED, BEL, AUT, COL) are upweighted by 2–3 percentage points over classical MC, and the top favourites (FRA, ESP, BRA) reduced by a commensurate amount. The encoding scales to roughly 25 qubits on a single GPU via cuQuantum, and is a natural candidate for the late NISQ window and the 50–100 logical-qubit regime expected from the first generations of fault-tolerant hardware.

References

- [1] Y. Aharonov, L. Davidovich, and N. Zagury. Quantum random walks. *Physical Review A*, 48:1687–1690, 1993.
- [2] Anonymous. The outcome prediction method of football matches by the quantum neural network based on deep learning. *Scientific Reports (Nature)*, June 2025. QNN classification task on European football leagues 2008–2022, not bracket-structural estimation.
- [3] Kishor Bharti, Alba Cervera-Lierta, Thi Ha Kyaw, Tobias Haug, Sumner Alperin-Lea, Abhinav Anand, Matthias Degroote, Hermann Heimonen, Jakob S. Kottmann, Tim Menke, et al. Noisy intermediate-scale quantum algorithms. *Reviews of Modern Physics*, 94:015004, 2022.
- [4] Ulrik Brandes, Gordana Marmulla, and Ivana Smokovic. Stop simulating! efficient computation of tournament winning probabilities. *arXiv preprint*, 2025.
- [5] Gilles Brassard, Peter Høyer, Michele Mosca, and Alain Tapp. Quantum amplitude amplification and estimation. *Quantum Computation and Information (Contemporary Mathematics)*, 305:53–74, 2002.
- [6] Shouvanik Chakrabarti, Rajiv Krishnakumar, Guglielmo Mazzola, Nikitas Stamatopoulos, Stefan Woerner, and William J. Zeng. A threshold for quantum advantage in derivative pricing. *Quantum*, 5:463, 2021.
- [7] Andrew M. Childs, Richard Cleve, Enrico Deotto, Edward Farhi, Sam Gutmann, and Daniel A. Spielman. Exponential algorithmic speedup by a quantum walk. *Proceedings of the 35th Annual ACM Symposium on Theory of Computing*, 2003.

- [8] Anthony C. Constantinou and Norman E. Fenton. Towards smart-data : improving predictive accuracy in long-term football team performance. *Knowledge-Based Systems*, 124:93–108, 2017.
- [9] Mark J. Dixon and Stuart G. Coles. Modelling association football scores and inefficiencies in the football betting market. *Journal of the Royal Statistical Society : Series C (Applied Statistics)*, 46(2):265–280, 1997.
- [10] Daniel J. Egger, Ricardo García Gutiérrez, Jordi Cahué Mestre, and Stefan Woerner. Credit risk analysis using quantum computers. *IEEE Transactions on Computers*, 70(12):2136–2145, 2021.
- [11] Gregory S. Engel, Tessa R. Calhoun, Elizabeth L. Read, Tae-Kyu Ahn, Tomáš Mančal, Yuan-Chung Cheng, Robert E. Blankenship, and Graham R. Fleming. Evidence for wavelike energy transfer through quantum coherence in photosynthetic systems. *Nature*, 446(7137):782–786, 2007.
- [12] Jaime Faj, Ivy Peng, Jacob Wahlgren, and Stefano Markidis. Quantum computer simulations at warp speed : Assessing the impact of gpu acceleration. *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2023.
- [13] Paul Glasserman. Monte carlo methods in financial engineering. 2004. Reference textbook on MC methods, used as the classical baseline benchmark.
- [14] Lov K. Grover. A fast quantum mechanical algorithm for database search. In *Proceedings of the 28th Annual ACM Symposium on Theory of Computing (STOC)*, pages 212–219, 1996.
- [15] Robin Hanson. Combinatorial information market design. *Information Systems Frontiers*, 5(1):107–119, 2003.
- [16] P. J. Hore and Henrik Mouritsen. The radical-pair mechanism of magnetoreception. *Annual Review of Biophysics*, 45:299–344, 2016.
- [17] E. Joos and H. D. Zeh. The emergence of classical properties through interaction with the environment. *Zeitschrift für Physik B Condensed Matter*, 59:223–243, 1985.
- [18] Julia Kempe. Quantum random walks : an introductory overview. *Contemporary Physics*, 44(4):307–327, 2003.
- [19] Neill Lambert, Yueh-Nan Chen, Yuan-Chung Cheng, Che-Ming Li, Guang-Yin Chen, and Franco Nori. Quantum biology. *Nature Physics*, 9:10–18, 2013.
- [20] Ashley Montanaro. Quantum speedup of Monte Carlo methods. *Proceedings of the Royal Society A*, 471:20150301, 2015.
- [21] Martin B. Plenio and Susana F. Huelga. Dephasing-assisted transport : quantum networks and biomolecules. *New Journal of Physics*, 10:113019, 2008.
- [22] Renato Portugal. *Quantum Walks and Search Algorithms*. Springer, 2 edition, 2018.
- [23] John Preskill. Quantum computing in the NISQ era and beyond. *Quantum*, 2:79, 2018.
- [24] PulsQuant. `scripts.tournament.simulate_bracket` — classical monte carlo bracket simulator. Internal codebase, 2026. Baseline MC implementation : logistic Elo + best-of-N series probability, 10 000 sims per run.

- [25] Quantum Frontiers. The World Cup from a Quantum Perspective. Blog post, Caltech IQIM, July 2018. Outlines a qudit encoding for the FIFA World Cup bracket and flags Grover amplification as an open extension. No formal publication.
- [26] Nikitas Stamatopoulos, Daniel J. Egger, Yue Sun, Christa Zoufal, Raban Iten, Ning Shen, and Stefan Woerner. Option pricing using quantum computers. *Quantum*, 4:291, 2020.
- [27] Yohichi Suzuki, Shumpei Uno, Rudy Raymond, Tomoki Tanaka, Tamiya Onodera, and Naoki Yamamoto. Amplitude estimation without phase estimation. *Quantum Information Processing*, 19:75, 2020.
- [28] Max Tegmark. Importance of quantum decoherence in brain processes. *Physical Review E*, 61:4194–4206, 2000.
- [29] Justin Wolfers and Eric Zitzewitz. Prediction markets. *Journal of Economic Perspectives*, 18(2):107–126, 2004.